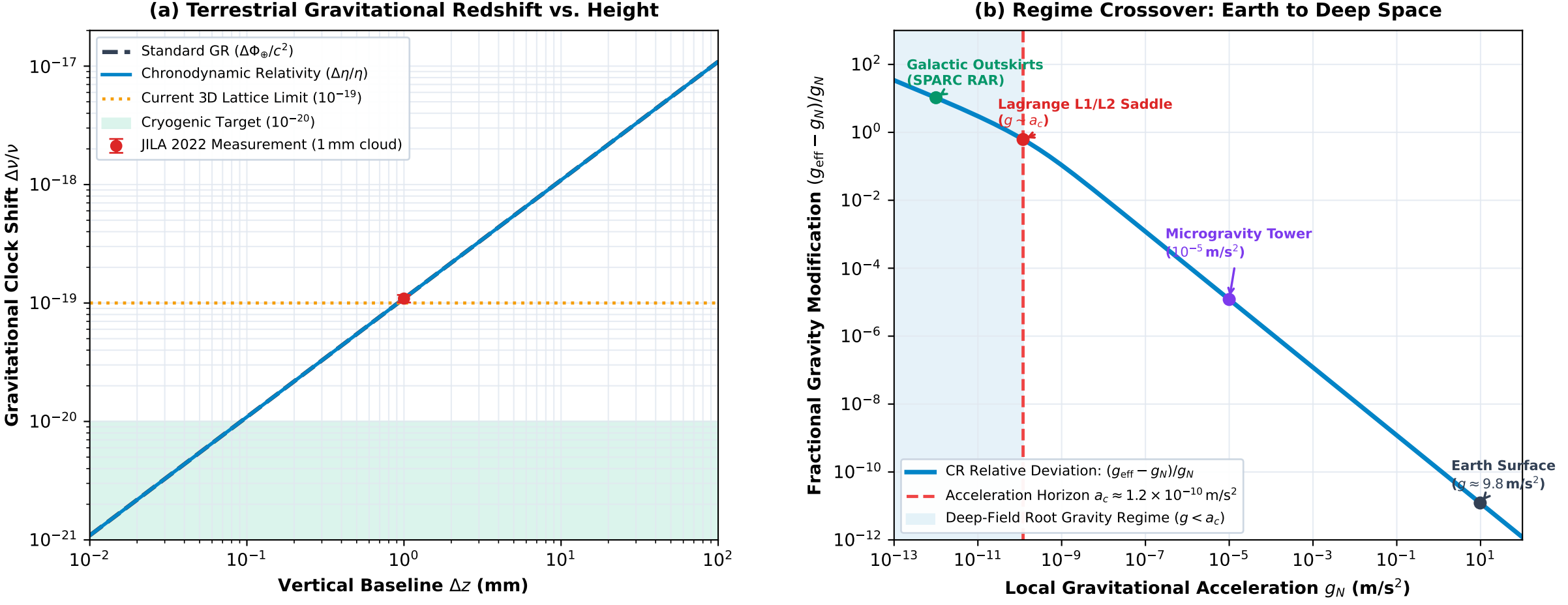
(a) Terrestrial Gravitational Redshift vs. Height
Standard GR ($\Delta\Phi_\oplus/c^2$)
Chronodynamic Relativity ($\Delta\eta/\eta$)
Current 3D Lattice Limit (10^{-19})
Cryogenic Target (10^{-20})
JILA 2022 Measurement (1 mm cloud)
Gravitational Clock Shift $\Delta\nu/\nu$
Vertical Baseline Δz (mm)
(b) Regime Crossover: Earth to Deep Space
Galactic Outskirts
(SPARC RAR)
Lagrange L1/L2 Saddle
($g \sim a_c$)
Microgravity Tower
(10^{-5} m/s^2)
Earth Surface
($g \approx 9.8$ m/s^2)
CR Relative Deviation: $(g_{\rm eff} - g_N)/g_N$
Acceleration Horizon $a_c \approx 1.2 \times 10^{-10}$ m/s^2
Deep-Field Root Gravity Regime ($g < a_c$)
Fractional Gravity Modification $(g_{\rm eff} - g_N)/g_N$
Local Gravitational Acceleration g_N (m/s^2)